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MT-302 Probability and Statistics

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TEXT BOOK AND MATERIALS

Probability and Statistics for Engineers and Scientists by Walpole, Myers, Myers and Ye

COURSE ASSESMENT

Quizzes	10%
Assignments	5%
Attendance	5%
Midterm Exams	30%
<u>Final Exam</u>	<u>50%</u>
Total	100%

What is Statistics

Specially to deal variation

The word “Statistics” which comes from the Latin words “status”. meaning a political state

The word statistics denotes a set of numerical data in the respective fields.

EXAMPLES

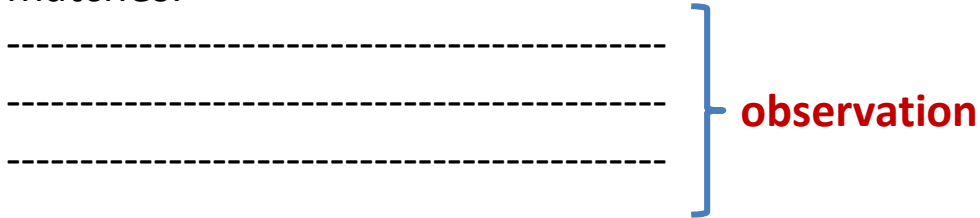
statistics of road accidents, statistics of crimes, statistics of births, statistics of educational institutions, etc.

EXAMPLES

Sachin Tendulkar

No of matches = 400

For example I get the score of 400 matches.



Average = 55.5 → **Statistic value**

We can say numerical facts that communicate information is **statistics**

Limitation of Statistics

1. Stats does not deal with the single value/number/figure .The statistic is a study of the group, not an individual.
2. Its results are approximated.
3. Results are true on long run.
4. Results are misleading.

Observation

Any numerically record information is called observation.

Example: 368 accidents in last two years

Constant

Those quantities which remains unchanged or fixed.

Example:

➤ No of days in a week = 7

➤ Value of $\pi = \frac{22}{7}$

Population

A **population** is the entire group that you want to draw conclusions about or the aggregate in which we are interested.

Examples:

Average height of Mechanical engineering students

Sample

A small part or subset of population is called sample.

Example:-

- Puff of perfume
- Few drops of blood

Types of Statistics

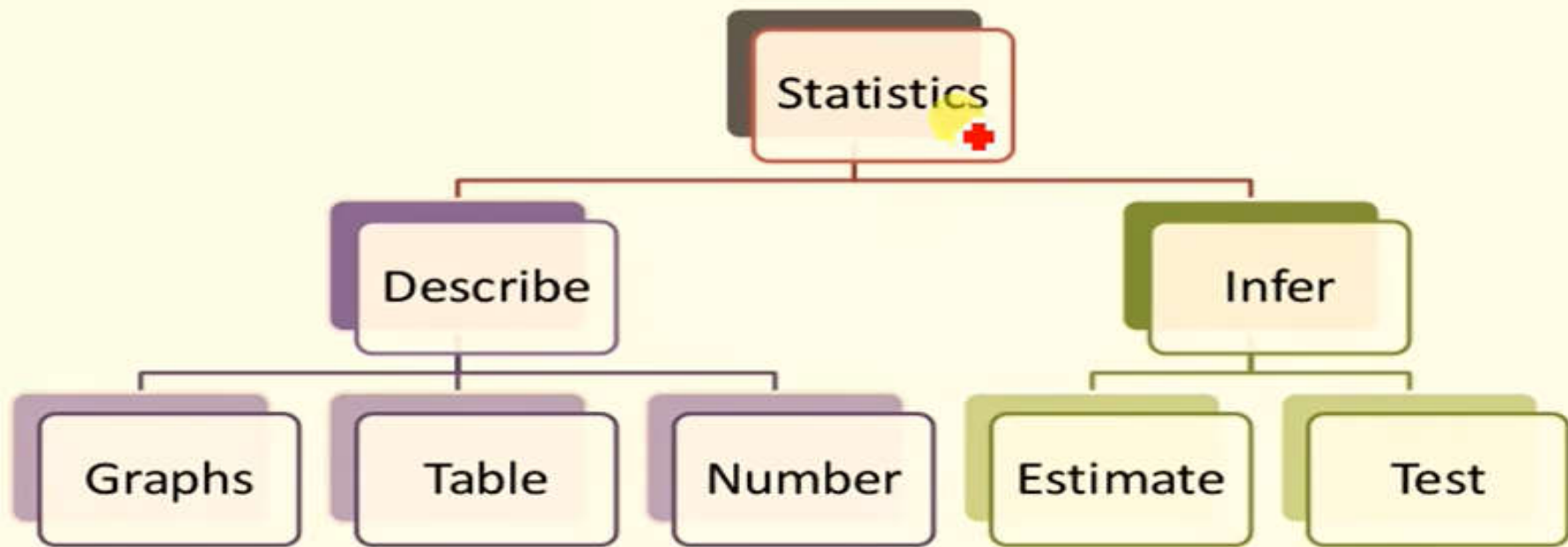
Descriptive Statistics

It consist of collection, organization, summarization, and presentation of data.

Inferential Statistics

The process of drawing results, conclusion about the population on the basis of sample evidence is called inferential statistics.

Statistics



EXAMPLES

1. Government college population = 1000 = N

We want to see how many students like statistics and how many do not

Take some samples say = n = 100

$$\text{Like} = \frac{60}{100} = 60\% \quad \text{and not like} = \frac{40}{100} = 40\%$$

Above result is based on sample if we use these results over the entire population so this will be inferential statistics.

2. MRI or CT scan.

3. Few drops of blood.

Discrete data

Discrete data typically only shows information for a particular event. Discrete data is a numerical type of data that includes whole, concrete numbers with specific and fixed data values determined by counting.

Examples of discrete data

Some discrete data examples are:

- ❖ The number of customers who bought different items
- ❖ The number of computers in each department
- ❖ The number of items you buy at the grocery store each week

What is continuous data

Continuous data is a type of quantitative data that represents precise measurements of nearly any numeric value. Often, a continuous data variable has many decimal points

Examples of continuous data

Some continuous data examples include:

1. The weight of newborn babies
2. The daily wind speed
3. The temperature of a freezer
4. Distance between university and home.

BASIS FOR COMPARISON	DISCRETE DATA	CONTINUOUS DATA
Meaning	Discrete data is one that has clear spaces between values.	Continuous data is one that falls on a continuous sequence.
Nature	Countable	Measurable
Values	It can take only distinct or separate values.	It can take any value in some interval.
Graphical Representation	Bar Graph	Histogram
Tabulation is known as	Ungrouped frequency distribution.	Grouped frequency distribution.
Classification	Mutually Inclusive	Mutually Exclusive
Function graph	Shows isolated points	Shows connected points
Example	Days of the week	Market price of a product

Difference Between Ungrouped Data and Grouped Data

1. Data that is gathered for the first time during a study or experiment, is called Ungrouped data.
2. ungrouped data or raw data is a mere list of numbers that does not convey anything.
3. Ungrouped data have neither been summarized nor organized in any way.
4. Grouping of data plays a significant role when we have to deal with large data.

Example:

Let ages of 12 wealthiest people in the world (just assume the data)
35, 29, 35, 24, 26, 29, 35, 33, 31, 28, 25, 24,

Data available in such a form is called **raw data**. And each entry i.e. 35, 34, 28, and so forth, is the **value** or **observation**.

The data presented in the above form does not provide sufficient information, such as the highest or lowest age. So, let us arrange this data in increasing order.

24, 24, 25, 26, 28, 29, 29, 31, 33, 35, 35, 35

When the number of observations is large, it becomes monotonous and time-consuming. Further, for easy understandability of the data, we will present it in tabular form i.e. **frequency distribution table for raw data or ungrouped data**.

Frequency distribution table for ungrouped data

Age	Tally Marks	Frequency
24	II	2
25	I	1
26	I	1
28	I	1
29	II	2
31	I	1
33	I	1
35	III	3

DATA IS IN ASCENDING ORDER

24, 24, 25, 26, 28, 29, 29, 31, 33, 35, 35, 35

Here, the number of times each number occurs is called **frequency**.

Frequency Distribution

A frequency distribution is the arrangement of data in tabular form which represents the frequency, with which each consecutive value of the variable occurs.

Grouped Data

When raw data have been grouped in different classes then it is said to be grouped data

When we have a large set of quantitative data, then we use group frequency distribution. It's useful to organize it into smaller intervals or classes and count how many data values fall into each class.

A frequency table does just that.

Frequency Table

Procedure:

HOW TO MAKE A FREQUENCY TABLE

1. Determine the number of classes and the corresponding class width.
2. Create the distinct classes. We use the convention that the *lower class limit* of the first class is the smallest data value. Add the class width to this number to get the *lower class limit* of the next class.
3. Fill in *upper class limits* to create distinct classes that accommodate all possible data values from the data set.
4. Tally the data into classes. Each data value should fall into exactly one class. Total the tallies to obtain each *class frequency*.
5. Compute the *midpoint* (class mark) for each class.
6. Determine the *class boundaries*.

Practice Problem #1

The following franchise fee data (in \$1000) was collected from 30 randomly selected franchises. Make a frequency table for these data using 6 classes

10	15	15	18	20	20	20	21	24	25	23	25	26	25	24
30	35	38	26	19	40	44	37	17	24	21	32	38	41	20

10 15 15 18 20 20 20 21 24 25 23 25 26 25 24
30 35 38 26 19 40 44 37 17 24 21 32 38 41 20

Step 1. First decide the number of classes. $\sqrt{n} = \sqrt{30} = 5.4 \approx 6$

here we have 6 classes

Step 2. Next, find the *class width*.

HOW TO FIND THE CLASS WIDTH (INTEGER DATA)

1. Compute
$$\frac{\text{Largest data value} - \text{smallest data value}}{\text{Desired number of classes}}$$
2. Increase the computed value to the next highest whole number.

To find the class width, we observe that the largest value is 44 and the smallest value is 10. Using 6 classes

$$\text{Class width} = \frac{44 - 10}{6} = 5.6 \approx 6 \quad (\text{Increase to } 6)$$

10	15	15	18	20	20	20	21	24	25	23	25	26	25	24
30	35	38	26	19	40	44	37	17	24	21	32	38	41	20

Step 3. Create the distinct classes. Determine the class limit for each class. The *lower class limit* is the lowest data value that can fit in a class. The *upper class limit* is the highest data value that can fit in a class.

The smallest value = 10.

Lower class limit of the first class = 10.

Class width = 6, we add 6 to 10 we get 16

Lower class limit for the *second* class = 16.

Lower class limit for 3rd class = $16+6 = 22$

Similarly class limit for 4th, 5th and 6th class are 28, 34 & 40 respectively.

Following above pattern, we establish all the lower class limits

Step 4.

Upper Class Limits.

Example 1 – *Solution*

cont'd

upper class limit = the lower class limit of that class + class width

$$1^{st} \text{ upper class limit} = 10 + 6 = 16$$

$$3^{rd} \text{ upper class limit} = 22 + 6 = 28$$

$$2^{nd} \text{ upper class limit} = 16 + 6 = 22$$

$$4^{th} \text{ upper class limit} = 28 + 6 = 34$$

$$5^{th} \text{ upper class limit} = 34 + 6 = 40$$

$$6^{th} \text{ upper class limit} = 40 + 6 = 46$$

Class Limits Lower - Upper
10 - 16
16 - 22
22 - 28
28 - 34
34 - 40
40 - 46

Step 5.

Example 1 – *Solution*

Next, tally the data into the six classes and find the frequency for each class.

•Procedure:

HOW TO TALLY DATA

Tallying data is a method of counting data values that fall into a particular class or category.

To tally data into classes of a frequency table, examine each data value. Determine which class contains the data value and make a tally mark or vertical stroke (I) beside that class. For ease of counting, each fifth tally mark of a class is placed diagonally across the prior four marks (||||).

The *class frequency* for a class is the number of tally marks corresponding to that class.

10 15 15
26 30 32

17 19 18 20 20 20 20 21 21
35 37 38 38

23 24 24 24 25 25 25 26
40 41 44

Class Limits Lower - Upper	Tally	Frequenc y
10 - 16		3
16 - 22	\	9
22 - 28	\	9
28 - 34		2
34 - 40		4
40 - 46		3

We want to count how many points are between 10 and 16 **NOT INCLUDING 16**

The right hand endpoint of any group isn't included in that group. It goes in the next group. That means 16 would be in the second group and any 22 we have would be counted in the third group.

Example 1 – *Solution*

Step 6 . Find the midpoints.

- The center of each class is called the *midpoint* (or *class mark*). The midpoint is often used as a representative value of the entire class.
- The midpoint is found by adding the lower and upper class limits of one class and dividing by 2.

$$\text{Midpoint} = \frac{\text{Lower class limit} + \text{Upper class limit}}{2}$$

Step 7 . Find the class boundaries

We use the following steps to calculate the class boundaries in a frequency distribution:

1. Subtract the upper class limit for the first class from the lower class limit for the second class.
2. Divide the result by two.
3. Subtract the result from the lower class limit and add the result to the upper class limit for each class.

The following examples show how to use these steps in practice to calculate class boundaries in a frequency distribution

Example 1 – *Solution*

Class Limits Lower - Upper
10 - 16
16 - 22
22 - 28
28 - 34
34 - 40
40 - 46

The upper class limit for the first class is 10 and the lower class limit for the second class is 16. Thus, we get: $31 - 30 = \mathbf{1}$.

Class Limits Lower – Upper	Class Boundaries Lower -Upper	Class Midpoint	Tally	Frequency
10 - 16	9.5 - 16.5	13		3
16 - 22	16.5 - 22.5	19	 	9
22 - 28	22.5 - 28.5	25	 	9
28 - 34	28.5 - 34.5	31		2
34 - 40	34.5 - 40.5	37		4
40 - 46	40.5 - 46.5	43		3

Total frequency : 30

Class Limits Lower - Upper	Class Boundaries Lower -Upper	Class Midpoint	Tally	Frequency	Relative frequency
10 - 16	9.5 - 16.5	13		3	$\frac{3}{30} = 0.1$
17 - 23	16.5 - 23.5	20		10	$\frac{10}{30} = 0.33$
24 - 30	23.5 - 30.5	27		9	$\frac{9}{30} = 0.3$
31 - 37	30.5 - 37.5	34		3	$\frac{3}{30} = 0.1$
38 - 44	37.5 - 44.5	41		5	$\frac{5}{30} = 0.16$

Total frequency : 30